

ANGERMUENDE, Germany

WMO: 102910

Lat: 53.0317N

Lon: 13.9908E

Elev: 182

StdP: 14.60

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.4	14.5	3.7	6.7	10.2	10.0	9.2	15.0	30.0	45.1	26.8	41.5	4.8	70	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	84.4	66.8	80.7	65.2	77.5	63.8	68.7	80.4	66.8	76.9	65.0	74.5	8.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	92.5	71.7	63.0	86.8	70.4	61.2	81.2	69.4	33.0	80.9	31.4	76.9	30.0	74.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	21.9	19.1	6.5	91.2	11.3	4.5	-1.6	94.5	-8.2	97.1	-14.5	99.6	-22.7	102.9		
(5)			5.8	71.8	10.9	1.8	-2.0	73.1	-8.4	74.2	-14.6	75.2	-22.5	76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	33.1	32.1	39.2	45.9	54.7	60.4	64.3	64.2	57.3	48.8	39.2	34.9		
	DBStd	13.47	9.88	8.95	7.18	6.46	6.13	5.70	5.30	5.70	4.99	6.70	6.36	7.09		
	HDD50	2448	525	503	339	158	24	0	0	0	3	101	326	469		
	HDD65	6383	990	922	801	575	322	160	79	86	238	502	775	933		
	CDD50	1690	0	1	3	33	170	313	443	440	221	65	1	0		
	CDD65	149	0	0	0	0	3	23	56	61	6	0	0	0		
	CDH74	1551	0	0	0	3	59	263	590	571	63	2	0	0		
	CDH80	383	0	0	0	0	1	55	145	176	6	0	0	0		
(14) Wind	WSAvg	8.9	11.3	9.4	9.6	8.4	7.5	7.6	7.7	7.3	8.3	8.8	9.9	10.8		
(15) Precipitation	PrecAvg	21.8	1.7	1.3	1.5	1.1	2.1	2.3	2.7	2.4	2.0	1.5	1.6	1.7		
	PrecMax	30.0	3.5	3.2	2.8	2.6	4.4	5.9	8.6	6.5	4.8	3.5	3.9	3.2		
	PrecMin	16.1	0.0	0.2	0.4	0.1	0.6	0.6	0.6	1.0	0.6	0.5	0.1	0.2		
	PrecStd	3.2	0.9	0.7	0.7	0.6	0.9	1.3	1.7	1.1	0.9	0.8	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.5	56.6	62.2	72.2	78.8	86.2	88.9	91.0	79.9	69.9	55.6	53.1		
		MCWB	48.3	49.7	50.6	55.3	62.4	65.5	68.9	69.9	64.2	60.4	50.9	48.5		
	2%	DB	48.6	51.0	55.9	66.1	75.6	81.2	83.7	84.8	75.2	64.9	52.6	50.7		
		MCWB	45.5	46.4	48.2	53.6	60.1	64.3	67.1	67.1	62.7	57.6	49.7	47.7		
	5%	DB	46.6	46.4	51.9	61.8	72.2	77.0	80.5	80.1	70.9	61.4	50.2	48.4		
		MCWB	43.6	42.5	45.8	51.7	58.6	62.5	65.5	65.1	60.3	55.4	47.5	45.8		
	10%	DB	44.3	42.9	48.9	57.5	68.3	73.1	77.0	76.0	66.7	58.3	47.9	45.7		
		MCWB	41.5	39.7	44.2	49.3	56.8	60.8	64.6	63.5	58.3	53.3	45.7	43.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	50.7	52.2	56.9	64.6	69.6	71.7	70.9	66.6	62.1	52.4	49.5		
		MCDB	51.4	55.2	60.2	68.2	75.1	77.5	82.9	87.3	77.2	68.2	54.0	51.6		
	2%	WB	46.1	46.1	49.1	54.5	61.8	66.1	69.0	68.4	63.8	58.7	50.1	48.0		
		MCDB	48.2	50.0	54.6	64.7	72.4	76.9	80.6	80.5	71.9	63.9	52.2	50.4		
	5%	WB	43.9	43.1	46.5	52.1	59.6	63.9	66.9	66.5	61.8	56.3	47.7	45.9		
		MCDB	46.3	46.2	50.8	60.3	70.1	73.9	77.3	76.4	68.8	60.3	49.7	48.0		
	10%	WB	41.7	40.0	44.3	49.9	57.6	61.9	64.9	64.8	59.7	53.9	45.9	43.4		
		MCDB	44.2	42.5	48.5	56.9	66.9	71.2	74.4	74.0	65.5	57.4	47.7	45.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	10.2	12.4	16.2	18.4	17.4	18.0	17.9	16.0	13.3	8.8	7.4		
		MCDBR	8.4	12.9	17.9	22.5	25.9	25.4	26.0	25.7	22.8	17.3	10.4	9.8		
	5% WB	MCWBR	7.6	10.0	11.9	12.9	14.1	12.2	12.2	11.4	12.4	11.1	8.6	8.9		
		MCWBR	8.6	13.1	16.0	21.0	23.0	20.9	23.1	21.5	19.7	15.6	10.2	9.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.343	0.378	0.412	0.409	0.407	0.427	0.421	0.384	0.366	0.342	0.312		
		taud	2.319	2.307	2.238	2.210	2.260	2.293	2.250	2.279	2.359	2.375	2.366	2.327		
	Ebn at Noon	Ebn at Noon	207	239	252	259	267	268	259	254	249	227	196	187		
		Edh at Noon	19	26	35	40	40	39	40	37	30	25	19	16		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	194	410	766	1286	1592	1683	1571	1345	981	530	231	138		
	RadStd	RadStd	22	64	92	177	189	135	208	116	112	71	30	13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page